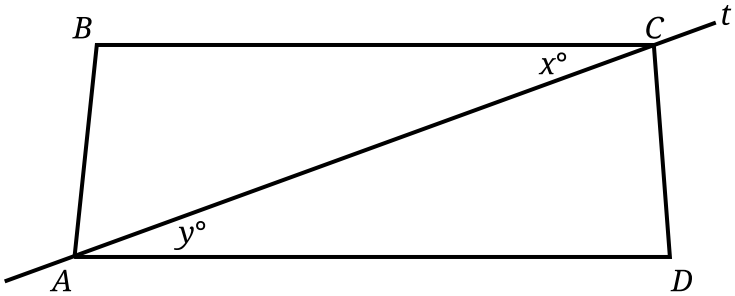


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure shown, $\overline{BC}$ is parallel to $\overline{AD}$ and $x = 22$. Line t passes through points A and C . What is the value of y ?

Answer

- A. 22
- B. 44
- C. 68
- D. 158

Correct Answer: A

Rationale

Choice A is correct. It's given that $\overline{BC}$ is parallel to $\overline{AD}$. Since line t intersects $\overline{BC}$ and $\overline{AD}$, it's a transversal. The angles in the figure marked as x° and y° are on opposite sides of the transversal and lie between the parallel segments, so they are alternate interior angles. When two parallel segments are intersected by a transversal, alternate interior angles have equal measure. It's also given that $x = 22$. Therefore, the value of y is 22.

Choice B is incorrect. This is the value of $2x$, not the value of y .

Choice C is incorrect. This is the complement of 22, not the value of y .

Choice D is incorrect. This is the supplement of 22, not the value of y .